Numericals on expansion waves

Problem:

A supersonic flow with $M_1 = 2$, $p_1 = 0.7$ atm, and $T_1 = 350$ K is expanded around a sharp corner through a deflection angle of 23◦. Calculate M_2, p_2, T_2, p_{02}, T_{02}, and the angles that the forward and rearward Mach lines make with respect to the upstream flow direction.

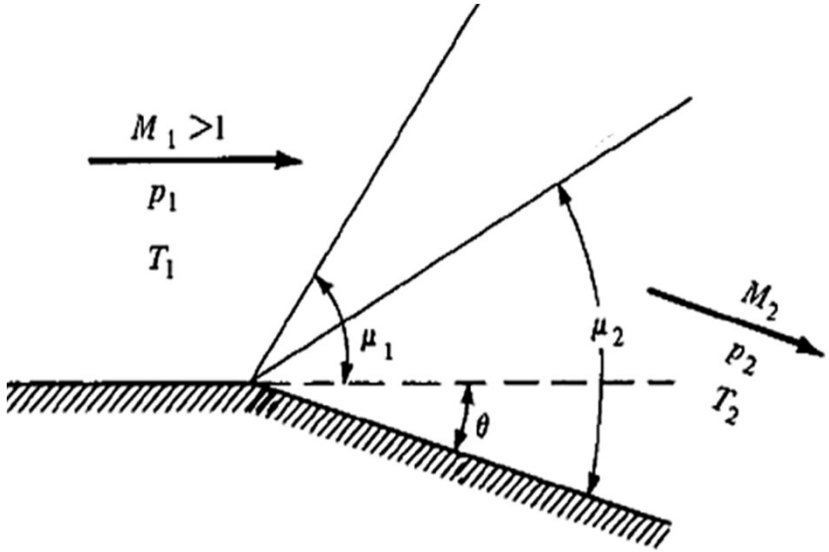
$$\text{For } \mathrm{M}_1 = 2 \xrightarrow{\text{from PM tables}} \nu(M_1) = 26.38$$

$$\theta = \nu(M_2) - \nu(M_1)$$

$$\Rightarrow \quad \nu(M_2) = \nu(M_1) + \theta$$

$$= 26.38 + 23$$

$$= 49.38$$



$$\text{For } \nu(M_2) = 49.38 \xrightarrow{\text{from PM tables}} M_2 = 2.98$$

For $M_1 = 2 \xrightarrow{\text{from isentropic tables}} \frac{p_1}{p_{01}} = 0.128 \quad \frac{T_1}{T_{01}} = 0.555$

$$\Rightarrow \quad p_{01} = 5.47 atm \quad T_{01} = 630.6 K$$

For $M_2 = 2.98 \xrightarrow{\text{from isentropic tables}} \frac{p_2}{p_{02}} = 0.0281 \quad \frac{T_2}{T_{02}} = 0.360$

Since $p_{01} = p_{02}$ and $T_{01} = T_{02}$

$$p_2 = \frac{p_2}{p_{02}} \times \frac{p_{01}}{p_1} \times p_1 = 0.0281 \times \frac{1}{0.128} \times 0.7$$

$$= 0.154 \text{ atm}$$

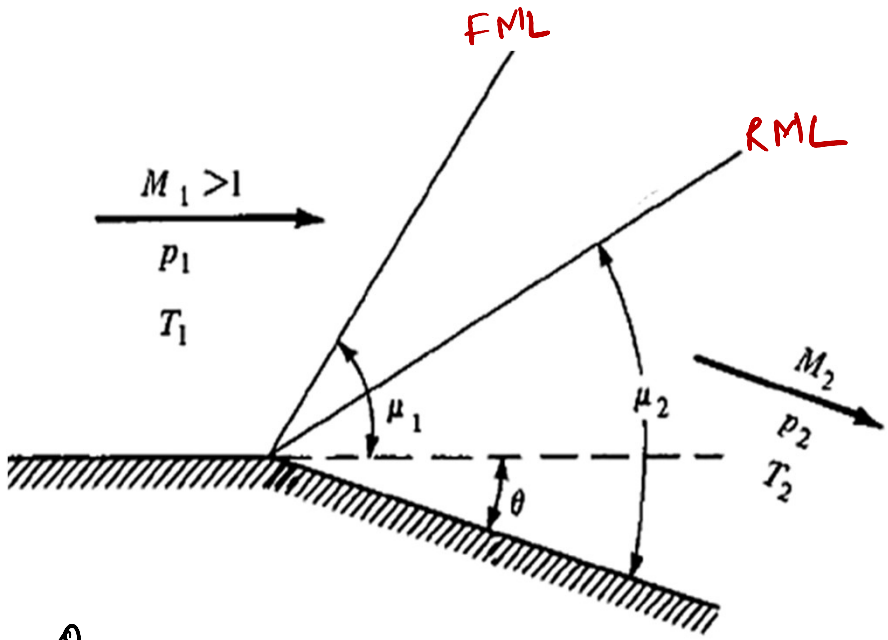
$$T_2 = \frac{T_2}{T_{02}} \times \frac{T_{01}}{T_1} \times T_1 = 0.36 \times \frac{1}{0.555} \times 350 = 227 \text{ K}$$

$$p_{02} = p_{01} = 5.47 \text{ atm}$$

$$T_{02} = T_{01} = 630.6 \text{ K}$$

the angles that the forward and rearward Mach lines make with respect to the upstream flow direction

- FML makes angle μ_1 w.r.t M_1
- RML makes an angle μ_2 w.r.t M_2



FML w.r.t M_1 is $\mu_1 = \sin^{-1}\left(\frac{1}{M_1}\right)$

$= \sin^{-1}\left(\frac{1}{2}\right)$

$=$

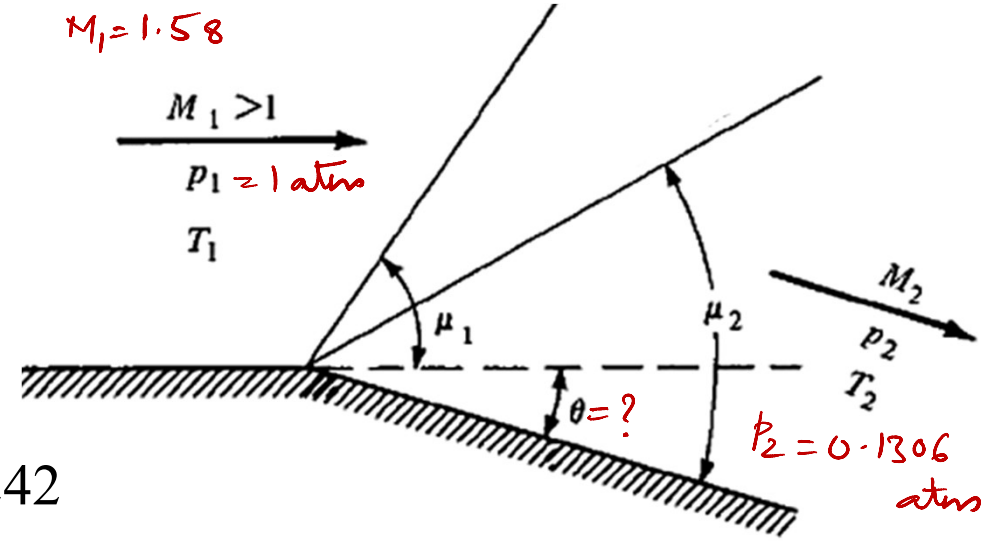
RML w.r.t M_1 is $\mu_2 - \theta = \sin^{-1}\left(\frac{1}{M_2}\right) - \theta$

$= \sin^{-1}\left(\frac{1}{2.98}\right) - 23^\circ \quad =$

Problem:

A supersonic flow with M_1 = 1.58, p_1 = 1 atm is expanded around a sharp corner. If the pressure after expansion is p_2 = 0.1306atm, find the deflection angle of the expansion corner.

$$\theta = \nu(M_2) - \nu(M_1)$$



For $M_1 = 1.58 \xrightarrow{\text{from PM tables}} \nu(M_1) = 14.268$

For $M_1 = 1.58 \xrightarrow{\text{from isentropic tables}} \dfrac{p_1}{p_{01}} = 0.242$

$$\Rightarrow \quad p_{01} = \frac{1}{0.242} = 4.13 atm$$

Since the expansion is isentropic, $\Rightarrow \quad p_{01} = p_{02}$

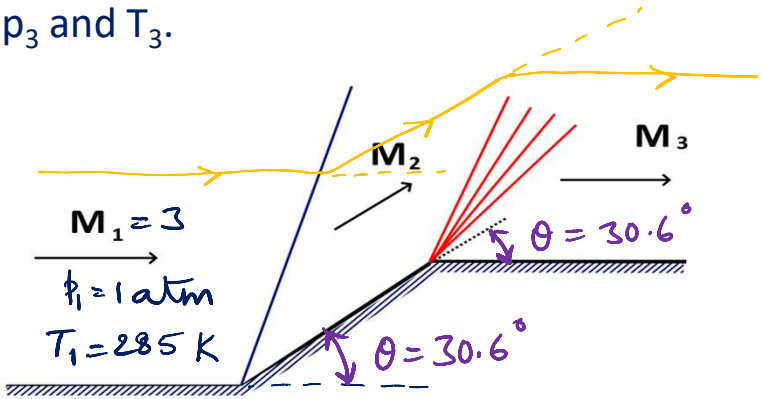
For $\dfrac{p_2}{p_{02}} = 0.0316 \xrightarrow{\text{from isentropic tables}} M_2 = 2.9$

For $M_2 = 2.9 \xrightarrow{\text{from PM tables}} \nu(M_2) = 47.79$

$$\begin{aligned}\theta &= \nu(M_2) - \nu(M_1) \\ &= 47.79 - 14.27 \\ &= 33.52^0\end{aligned}$$

Problem:

Consider the supersonic flow over a compression corner and an expansion corner as shown here. If M_1=3, p_1=1 atm and T_1=285K, the flow turning angle is 30.6^0, find M_3, p_3 and T_3.



1-2: Oblique shock

$$\text{For } M_1 = 3 \text{ and } \theta = 30.6^0 \xrightarrow{\text{From } \theta-\beta-M \text{ chart}} \beta = 53^0$$

$$M_{n1} = M_1 \sin\beta = 3\sin 53^0 = 2.4$$

$$\text{For } M_{n1} = 2.4 \xrightarrow{\text{From } NS \text{ tables}} M_{n2} = 0.523,\ \frac{p_2}{p_1} = 6.553,\ \frac{T_2}{T_1} = 2.04,\ \frac{p_{02}}{p_{01}} = 0.54$$

$$M_2 = \frac{M_{n2}}{\sin(\beta-\theta)} = \frac{0.523}{\sin(53-30.6)} = 1.37$$

For $M_1 = 3 \xrightarrow{\text{from isentropic tables}} \frac{p_1}{p_{01}} = 0.0272 \quad \frac{T_1}{T_{01}} = 0.357 \quad \Rightarrow p_{01} = \frac{p_1}{0.0272} = 36.8 atm$

$$T_{01} = \frac{T_1}{0.357} = 798.3K$$

$$\frac{p_{02}}{p_{01}} = 0.54 \Rightarrow p_{02} = 0.54 \times p_{01} = 0.54 \times 36.8 = 19.87 atm$$

2-3: Prandtl-Meyer Expansion

For $\mathrm{M}_2 = 1.37 \xrightarrow{\text{from PM tables}} \nu(M_2) = 8.413$

$$\begin{aligned} \nu(M_3) &= \nu(M_2) + \theta \\ &= 8.413 + 30.6 \\ &= 39.013 \end{aligned}$$

For $\nu(M_3) = 39.01 \xrightarrow{\text{from PM tables}} M_3 = 2.5$

$$T_{01} = T_{02} = T_{03} = 798.3K$$

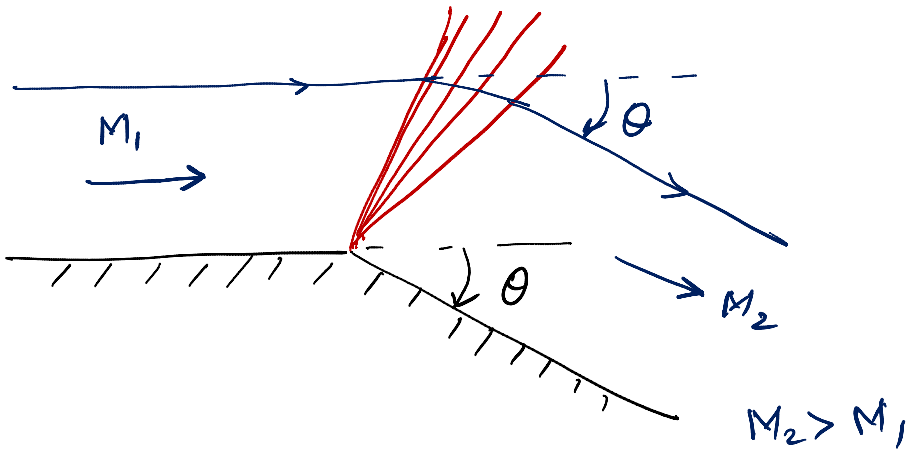
$$p_{02} = p_{03} = 19.87atm$$

For $M_3 = 2.5 \xrightarrow{\text{from isentropic tables}} \frac{p_3}{p_{03}} = 0.0585 \quad \frac{T_3}{T_{03}} = 0.444$

$$p_3 = 0.0585 \times p_{03} = 1.162\text{atm}$$

$$T_3 = 0.444 \times T_{03} = 354.4K$$

Maximum expansion angle for a Prandtl-Meyer expansion wave.



$$\boxed{\theta = \nu(M_2) - \nu(M_1)}$$

Where,

$$\nu(M) = \sqrt{\frac{\gamma+1}{\gamma-1}}\tan^{-1}\sqrt{\frac{\gamma-1}{\gamma+1}(M^2-1)} - \tan^{-1}\sqrt{M^2-1}$$

Maximum θ $\rightarrow$ Depends on M_1 and M_2

θ will be maximum for $M_1 = 1$ and $M_2 = \infty$

$$\nu(M_1 = 1) = 0$$

$$\nu(M_2 = \infty) = \left(\sqrt{\frac{\gamma+1}{\gamma-1}} - 1\right)90^\circ = 130.45^0$$

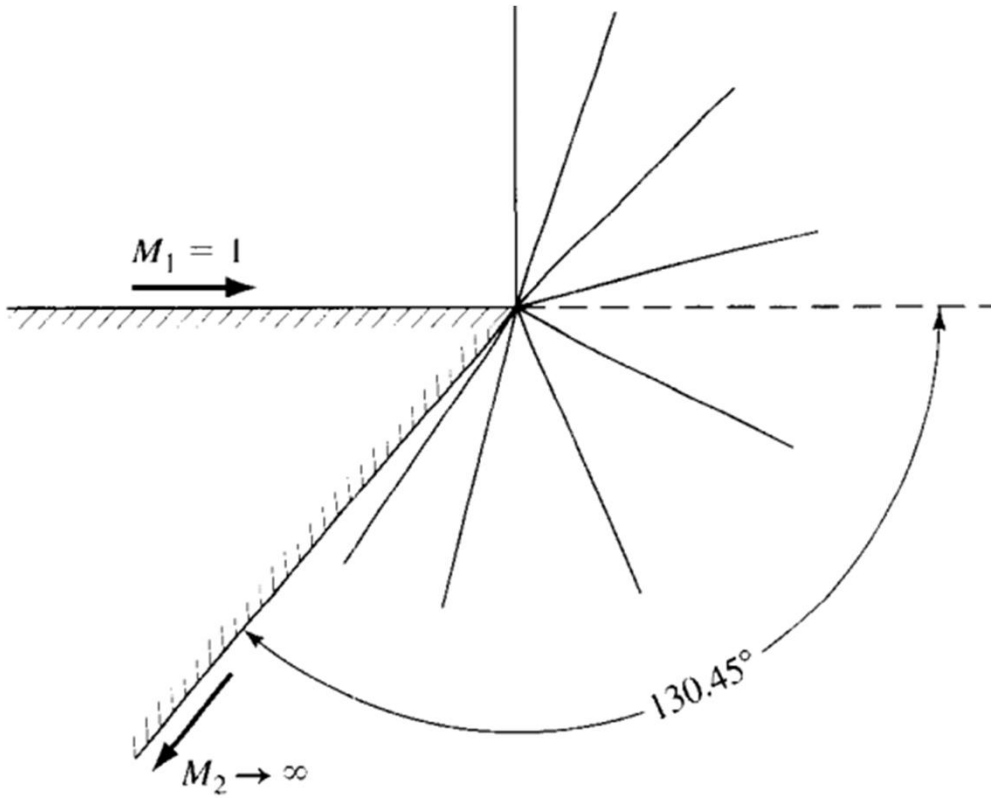


Figure 4.39 | Maximum expansion angle for a Prandtl–Meyer centered expansion wave.